

I2OS_GPC: Modular Market Response Operating System Architecture

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Abstract

This paper presents I2OS_GPC, a modular market response operating system designed to transform conventional Expert Advisor architectures into structure-driven execution systems. Unlike traditional trading systems based on prediction and indicator signals, I2OS_GPC executes trades only after multi-layer structural verification and real price movement confirmation. The architecture decomposes trading logic into independent modules, enabling precise control, interpretability, and suppression of false signals.

1 Introduction

Conventional automated trading systems rely on tightly coupled logic:

Indicator -> Condition -> Entry

This approach suffers from lack of structural separation, sensitivity to false signals, and limited interpretability.

I2OS_GPC introduces a modular operating system architecture:

Sensor -> Scanner -> Survival -> Reality -> Execution -> Collapse -> Memory

2 System Definition

The system is defined as:

$$\mathcal{GPC} = \{\mathcal{S}, \mathcal{C}, \mathcal{V}, \mathcal{R}, \mathcal{E}, \mathcal{X}, \mathcal{M}\} \quad (1)$$

3 Sensor Module

Market state is represented as:

$$S(t) = (F, \Sigma, E, \phi, P, E_{res}, Pr) \quad (2)$$

where F is Flow, Σ is Sync, E is Energy, ϕ is Phase, P is Pulse, E_{res} is Resonance, and Pr is Probability.

4 Scanner Module

Candidate detection is defined as:

$$C(t) = \mathbf{1}[F > \theta_F \wedge \Sigma > \theta_\Sigma \wedge E > \theta_E \wedge \phi < \theta_\phi] \quad (3)$$

5 Survival Module

Persistence verification is defined as:

$$V(t) = \mathbf{1}[S(t + \Delta t) \approx S(t)] \quad (4)$$

6 Reality Module

Real movement confirmation is defined as:

$$R(t) = \mathbf{1}[\Delta p_{fav} > \theta_{fav}] \quad (5)$$

with additional constraints:

$$\Delta F > \theta_{dF}, \quad P > \theta_P. \quad (6)$$

7 Execution Module

Execution is permitted only when:

$$E(t) = C(t) \wedge V(t) \wedge R(t) \wedge \neg Safety(t). \quad (7)$$

8 Collapse Module

Structural collapse is detected as:

$$X(t) = \mathbf{1}[\min(F, \Sigma, E) < \theta_X] \quad (8)$$

or when adverse movement exceeds the loss threshold.

9 Memory Module

State recording is defined as:

$$M(t + 1) = M(t) \cup \{S, C, V, R, E, X\}. \quad (9)$$

10 Experimental Observation

10.1 Experimental Setup

The system was evaluated using live MetaTrader execution logs from the I2OS_GPC v7.3N prototype. The experiment focused on whether the newly introduced Reality Module could suppress weak entries observed in v7.3M while still allowing valid GO executions.

The evaluated logs included:

- decision logs,
- execution-state logs,

- close logs,
- heartbeat logs,
- state snapshot logs.

The system was operated under the following principle:

Do not relax global thresholds. Verify whether the Reality Module can filter weak candidates before execution.

10.2 Observed Candidate Filtering

During the initial v7.3N observation period, the system produced many candidate states but almost no entries. This behavior was expected because v7.3N adds stricter initial-move confirmation.

The dominant rejection patterns were:

- no favorable movement,
- direction reversal during pending state,
- flow not rising,
- weak pulse,
- flow decay,
- freshness expiration.

These rejection reasons indicate that the system was not inactive. Instead, it was actively scanning candidates and rejecting those that lacked real favorable movement.

10.3 First Valid GO Observation

A valid GO was later observed under the v7.3N logic.

The trade characteristics were:

- direction: BUY,
- result: positive close,
- maximum favorable excursion: positive and clearly above the near-zero profile observed in weak v7.3M entries,
- exit reason: profit protection.

The GO occurred after the following structural transition:

candidate → survival → initial movement confirmation → GO execution

The observed GO condition included:

- increasing Flow,

- improved Phase,
- recovered Pulse,
- favorable price movement before entry.

This supports the hypothesis that the Reality Module improves GO quality by requiring actual price confirmation.

10.4 v7.3M vs v7.3N Behavioral Difference

The behavioral shift is summarized as follows:

Aspect	v7.3M	v7.3N
Entry trigger	structural alignment	alignment + real movement
Weakness	weak post-rebound entries	low frequency
Typical weak trade	near-zero max favorable movement	filtered before entry
Core improvement	entry recovery	reality verification
Observed behavior	more entries, weaker quality	fewer entries, higher selectivity

Table 1: Behavioral comparison between v7.3M and v7.3N.

10.5 Interpretation

The experimental logs indicate that v7.3N successfully changed the system from an entry-recovery model to a selective initial-move verification model.

The most important observation is not the number of trades. The important observation is that weak candidates were rejected before execution, while the first valid GO produced a favorable excursion and a profit-protection exit.

This suggests that the Reality Module acts as a false-signal suppressor rather than a mere threshold tightening mechanism.

10.6 Limitations

This experiment is preliminary and based on a limited observation window. The result should not be interpreted as statistical proof of profitability.

The current evidence supports only the following claims:

- the modular pipeline operates correctly,
- candidate rejection is explainable by module-level reasons,
- v7.3N suppresses weak entries more effectively than v7.3M,
- at least one valid GO was observed after reality confirmation.

Further evaluation requires longer live logs, multiple market sessions, and module-level comparison between winning and losing GO events.

10.7 Experimental Conclusion

The v7.3N experiment supports the architectural claim that market execution should not occur immediately after structural alignment.

Instead, execution should require:

$$Candidate \wedge Survival \wedge Reality \wedge SafetyClear$$

This confirms the central design principle of I2OS_GPC:

Enter only when structure survives and reality confirms movement.

11 Conclusion

I2OS_GPC transforms a trading EA into a modular market response OS.

Structure Detection

- + Survival Filtering
- + Reality Verification
- + Controlled Execution
- + Collapse Handling
- + Structural Memory

The central principle is:

Enter only when structure survives and reality confirms movement.

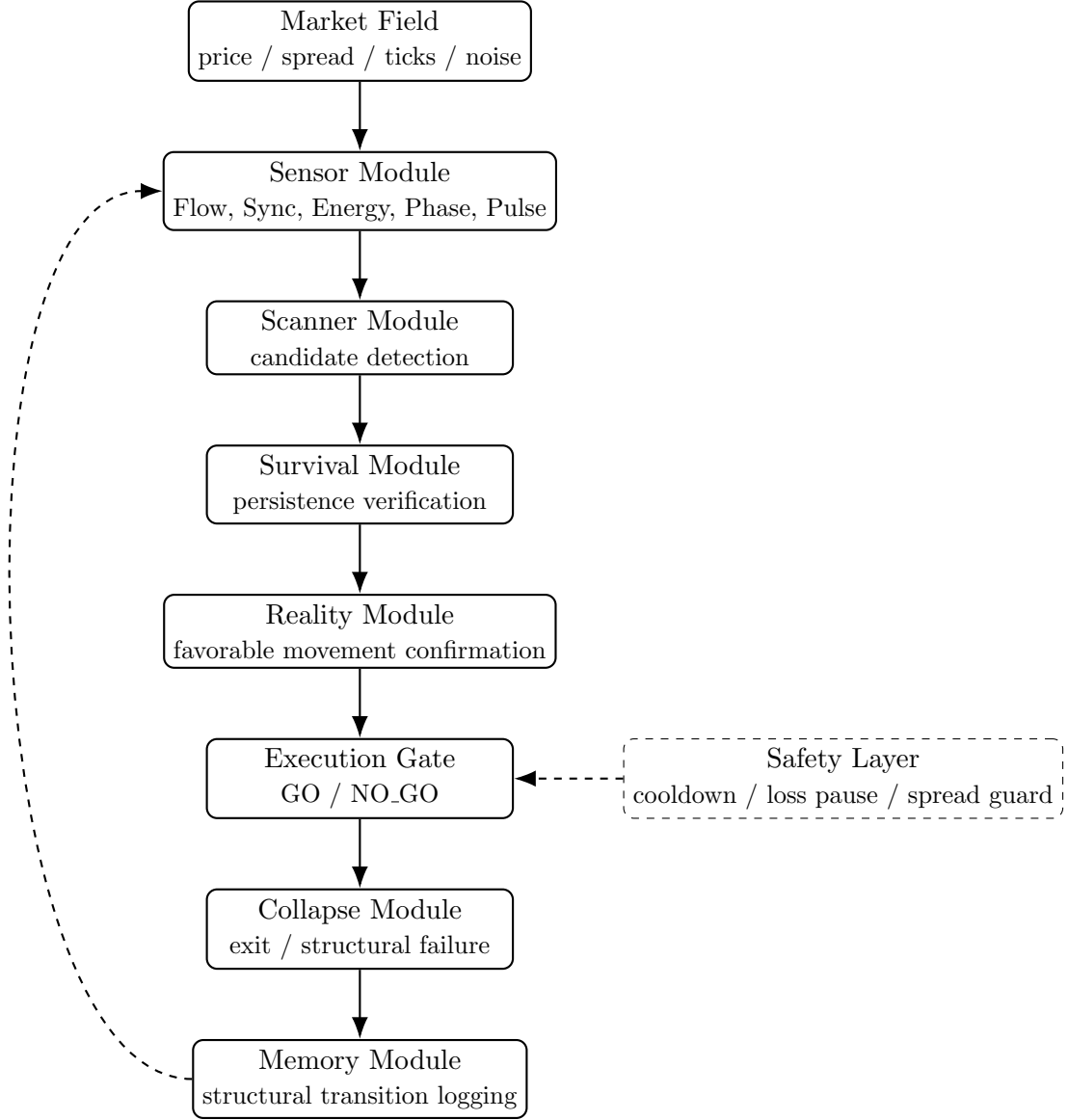


Figure 1: I2OS_GPC modular market response operating system. The architecture separates observation, candidate detection, persistence verification, reality confirmation, execution, collapse handling, and memory logging.

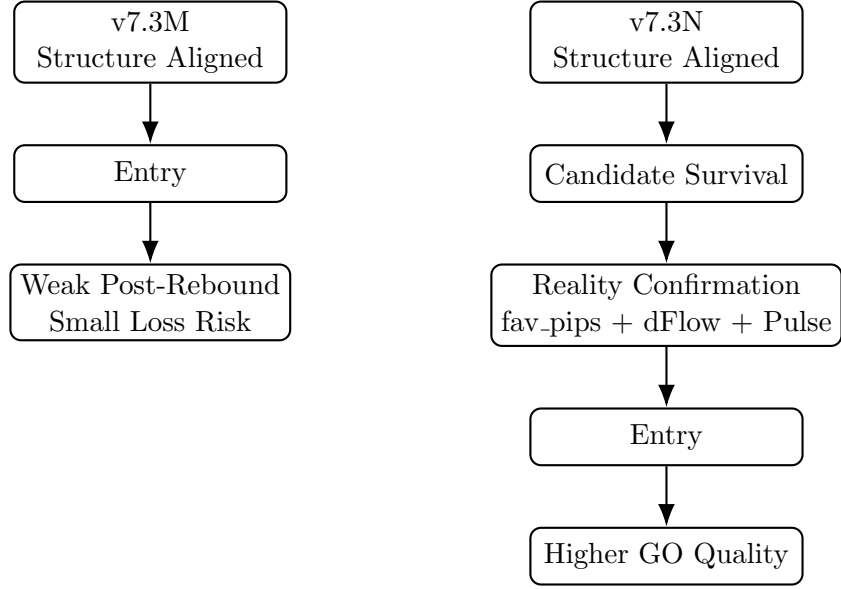


Figure 2: Execution logic transition from v7.3M to v7.3N. v7.3M executes after structural alignment, while v7.3N requires additional survival and reality confirmation before entry.

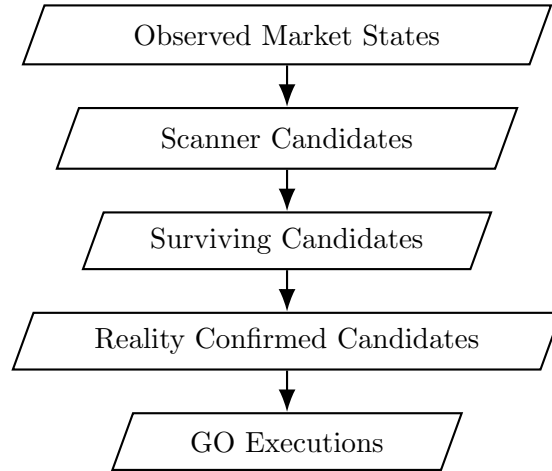


Figure 3: Candidate filtering funnel. The system intentionally reduces trade frequency by rejecting structures that fail persistence or reality confirmation.